

Can You Keep a Secret?

A Brief Introduction to Cryptography

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Wabash.

Outline

- 1 Cryptography
- 2 Symmetric Cryptography
 - Transposition Ciphers
 - Substitution Ciphers
- 3 Asymmetric Cryptography
 - Diffie-Hellman Key Exchange
 - RSA Encryption
 - Elliptic Curve Cryptography
- 4 Internet Communication

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Etymology

- κρυπτός (hidden, secret) from κρύπτω (hide, cover)
- γράφω (write, draw)

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Examples

- ASCII, Unicode
- Morse Code
- Universal Product Code (UPC)

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- Codes were more common before the invention of the telegraph.

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- Alice sends a message to Bob encrypted by a key only she and Bob knows.
- If Eve knows the key, she can decrypt the message.

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W . . . S . . . W . . . F . . . T .
• . A . A . H . L . A . S . I . H . S
 . . B . . . A . . . Y . . . G . . .

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- Cipher text is “WSWFTA AHLASIH SBAYG”.

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- Plaintext is written on successive rows of fixed length.
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Types

Simple Substitution Cipher: operates on single letters

Polygraphic Cipher: operates on groups of letters

Monoalphabetic Cipher: uses fixed substitution over entire message

Polyalphabetic cipher: uses different substitutions at different positions in the message

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- Cipher text is “ZDEDVKDOADBVLJKWV”.

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- Keyword to determine which alphabet used

Vigenère Cipher Example

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
A	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
B	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A
C	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B
D	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C
E	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D
F	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E
G	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F
H	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G
I	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H
J	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I
K	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J
L	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K
M	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L
N	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M
O	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N
P	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O
Q	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P
R	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q
S	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R
T	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S
U	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T
V	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U
W	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V
X	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W
Y	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X
Z	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y

Vigenère Cipher Example

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
A	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
B	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A
C	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B
D	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C
E	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D
F	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E
G	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F
H	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G
I	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H
J	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I
K	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J
L	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K
M	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L
N	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M
O	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N
P	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O
Q	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P
R	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q
S	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R
T	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S
U	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T
V	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U
W	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V
X	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W
Y	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X
Z	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y

Key: GOODRICHGOODRICHGO

Plain text: WABASHALWAYS FIGHTS

Cipher text: COPDJPCS COMVMQIOZG

Enigma Machine



- Set plugboard and rotors
- Press key on keyboard
- Rotors with randomly wired contacts rotate to new position
- Read letter from the lampboard

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Steps

- Put message in 4×4 column-major order matrix of bytes (the *state*)
- Expand a short key into a number of separate round keys
- AddRoundKey – Combine each byte of the state with the round key using bitwise XOR
- SubBytes – Replace each byte with another from lookup table
- ShiftRows – Shift last three rows cyclically some number of steps
- MixColumns – Combine the four bytes of each column

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- Security based on operations easy to perform, but hard to invert.
- One-way functions described in 1874 by William Jevons.

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 - Difficult to compute x given y .

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Example

- $p = 101, g = 7$.
- $a = 5 \implies A = 7^5 \bmod 101 = 41$

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- $p = 101, g = 7$.
- $a = 5 \implies A = 7^5 \bmod 101 = 41$
- $b = 11$

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- ➊ Alice and Bob agree on prime p and base g . (Both are public.)
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RSA Encryption

- ➊ Bob picks two distinct prime numbers p, q . (Private)
- ➋ Bob computes product $m = pq$. (Public)
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- ➏ Alice can encrypt a message M with $\gcd(M, m) = 1$ as $C = M^e \bmod m$.
- ➐ Bob decrypts the ciphertext C as $M = C^d \bmod m$.

RSA Encryption

Example

- $p = 101, q = 103$

RSA Encryption

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- $p = 101, q = 103 \implies m = 10403, \varphi(m) = 10200.$

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- Convert "WABASH ALWAYS FIGHTS" to ASCII:

RSA Encryption

Example

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- $e = 2243 \implies d = 5507$.
- Convert "WABASH ALWAYS FIGHTS" to ASCII:
 $M = [87, 65, 66, 65, 83, 72, 32, 65, 76, 87,$
 $65, 89, 83, 32, 70, 73, 71, 72, 84, 83]$

RSA Encryption

Example

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 $C = [2620, 1835, 5420, 1835, 6113, 2874, 544, 1835, 8507, 2620, 1835, 7335, 6113, 544, 1726, 848, 9447, 2874, 1198, 6113]$

RSA Encryption

Example

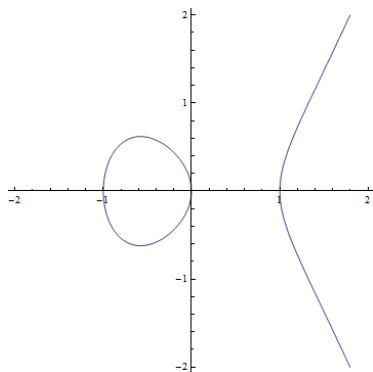
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Elliptic Curves

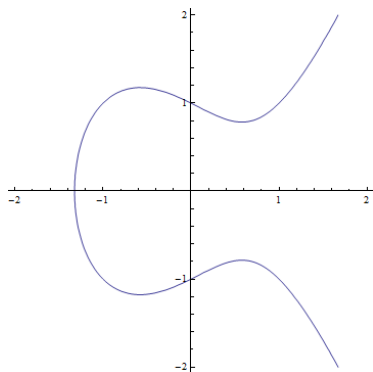
$$y^2 = x^3 + ax + b$$

Elliptic Curves

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$$y^2 = x^3 - x$$



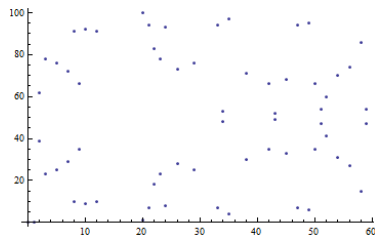
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Elliptic Curves in Finite Fields

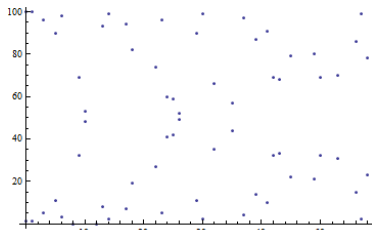
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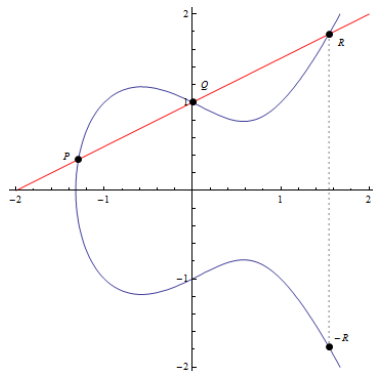
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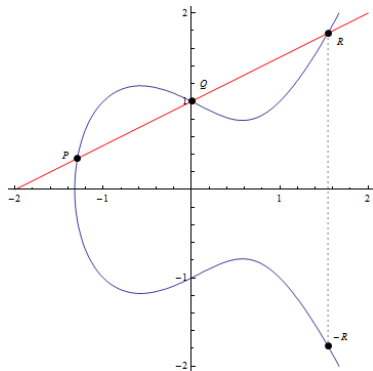


$$P + Q + R = 0$$

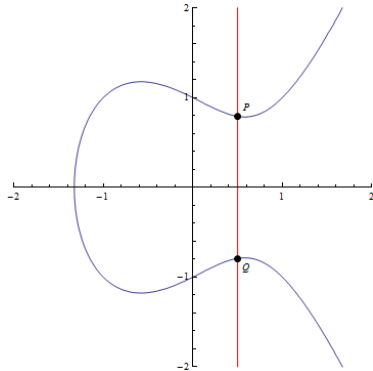
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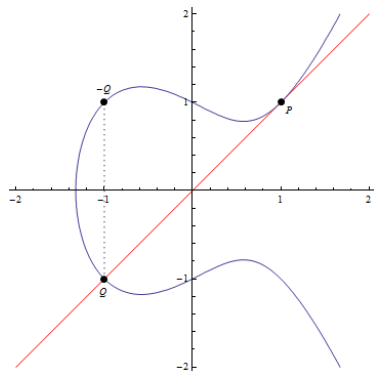


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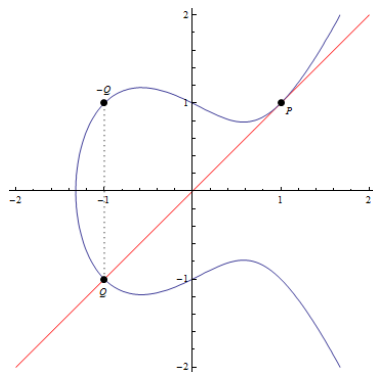
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Elliptic Curve Group Operation



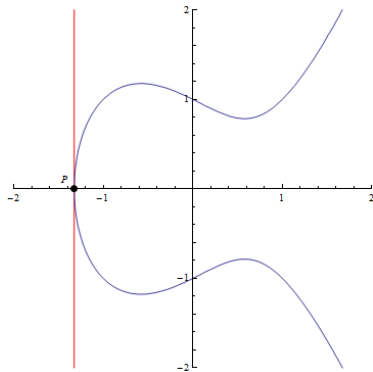
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Elliptic Curve Group Operation



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$$P + P + 0 = 0$$

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Elliptic Curve Diffie-Hellman

- Alice and Bob agree on public parameters:
 - Define elliptic curve p, a, b : $y^2 \equiv x^3 + ax + b \pmod{p}$
 - Point on curve G
- Alice picks private integer d_A with $0 \leq d_A < p$.
 - Alice sends Bob $Q_A = d_A G$.
- Similarly, Bob picks d_B and sends Alice $Q_B = d_B G$.
- Alice and Bob compute $(x_k, y_k) = d_A Q_B = d_B Q_A$.
 - Shared secret is x_k .

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- Authenticate certificates (Asymmetric)
 - Establish shared secret (Asymmetric)
 - Conduct session communication in secret (Symmetric)